# Exercise 9

# On vectors in the space

From the school book | Understand | Apply | Higher Order Thinking Skills

**1** Express the vectors $(-1, 0, 0)$ , $(0, -1, 0)$ , $(0, 0, -1)$ in terms of the fundamental unit vectors.

**2** **Find the norm of each of the following vectors :**

**( 1 )** $\vec{A} = (2, -1, 0)$

**( 2 )** $\vec{A} = (1, 2, -2)$

**( 3 )** $\vec{B} = 5\hat{i} - 4\hat{j} + 2\sqrt{2}\,\hat{k}$

**( 4 )** $\vec{C} = \hat{k}$

**( 5 )** $\vec{B} = 3\hat{i} + 4\hat{j}$

« $\sqrt{5}$ , 3 , 7 , 1 , 5 »

**3** If $\vec{A} = -3\hat{i} - 5\hat{j} + \hat{k}$ , $\vec{B} = 4\hat{j} + 8\hat{k}$ **Find :**

**( 1 )** $\vec{A} - \frac{1}{2}\vec{B}$ **( 2 )** $3\vec{B} - 2\vec{A}$ **( 3 )** $2\vec{A} + 3\vec{B}$

**4** If $\vec{A} = (2, -3, 1)$ , $\vec{B} = (4, -2, 0)$ , $\vec{C} = (-6, 0, 3)$

**Find each of the following vectors :**

**( 1 )** $\vec{A} + \vec{B}$ **( 2 )** $3\vec{A} - \frac{1}{3}\vec{C}$ **( 3 )** $\frac{3}{2}\vec{B} + \frac{2}{3}\vec{C}$

**5** If $\vec{C} = (2, -3, 1)$ , $\vec{D} = (0, 2, -2)$

**( 1 ) Find :** $5\vec{C} - 2\vec{D}$ **( 2 )** If $3\vec{A} - 4\vec{D} = \vec{C}$ , find : $\vec{A}$

**6** If $\vec{C} = (4, -1, 2)$ , $\vec{D} = (-3, 4, 5)$ , $\vec{E} = (-5, 1, 3)$

Find the vector $\vec{A}$ which satisfies the equation : $2\vec{A} = 2\vec{C} - 3\vec{D} + 2\vec{E}$

**7** **Choose the correct answer from the given answers :**

**( 1 )** If $\vec{A} = (-3, 7, 8)$ , then $\|\vec{A}\|$ = .....

(a) 12 (b) $\sqrt{222}$ (c) $\sqrt{122}$ (d) 10

**( 2 )** *(2nd Session 2019)* If $\vec{A} = (3, -2, m)$ , $\|\vec{A}\| = \sqrt{22}$ , then m = .....

(a) 21 (b) $\pm 9$ (c) $\pm 3$ (d) 17

**( 3 )** If $\vec{A} = (2, -3, 5)$ , $\vec{B} = (-2, 0, 4)$ , then $3\vec{A} - 2\vec{B}$ = .....

(a) (10 , −9 , 7) (b) (4 , −9 , 19) (c) (15 , 11 , 8) (d) (11 , 12 , 4)

**( 4 )** If $\vec{A} = (-1, 5, -2)$ , $\vec{B} = (3, 1, 1)$ and if $\vec{A} + \vec{B} + \vec{C} = \hat{i}$ , then $\vec{C}$ = .....

(a) $\hat{i} + 6\hat{j} - \hat{k}$ (b) $-\hat{i} - 6\hat{j} + \hat{k}$ (c) $\hat{i} + 4\hat{j} - 3\hat{k}$ (d) $\hat{i} + 4\hat{j} - \hat{k}$

**( 5 )** If $\vec{A} = (1, -1, 2)$ , $\vec{B} = (0, 2, -3)$ , $\vec{C} = (-2, 1, 0)$ , then $\| 3\vec{A} - \vec{B} + \vec{C} \|$ = .....

(a) $8\sqrt{3}$ (b) 11 (c) 12 (d) $7\sqrt{2}$

**( 6 )** If $\vec{A} = (2, 3\sqrt{3}, 0)$ , $\vec{B} = (4, 2\sqrt{3}, \sqrt{6})$ , then $\| \overrightarrow{AB} \|$ = .....

(a) 12 (b) 13 (c) $\sqrt{13}$ (d) 5

**( 7 )** *(1st Session 2019)* If $\overrightarrow{AB} = -3\hat{i} + 3\hat{j} + 7\hat{k}$ , $\overrightarrow{BC} = \hat{j} + 5\hat{k}$ , then $\| \overrightarrow{AC} \|$ = .....

(a) 13 (b) 12 (c) 10 (d) 9

**( 8 )** *(1st Session 2018)* If $\vec{A} = (-2, 4, 6)$ , $\vec{B} = (0, k, 3)$ where $k \in \mathbb{Z}^+$ and $\| \overrightarrow{AB} \| = 7$ , then the value of k = .....

(a) 10 (b) 8 (c) 6 (d) 4

**( 9 )** If $\| k(3, -3\sqrt{3}, 0) \| = 1$ , then k = .....

(a) $\frac{1}{6}$ (b) $\pm\frac{1}{6}$ (c) 6 (d) $\pm 6$

**(10)** If $\vec{A} = (-9, 1, 8)$ , $\vec{B} = (-6, -3, -4)$ , then the unit vector in direction of $\overrightarrow{AB}$ = .....

(a) $\left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$ (b) $\left(\frac{3}{13}, \frac{-4}{13}, \frac{-12}{13}\right)$

(c) $\left(\frac{-3}{13}, \frac{4}{13}, \frac{12}{13}\right)$ (d) $\left(\frac{-3}{13}, \frac{-4}{13}, \frac{-12}{13}\right)$

**(11)** The vector which represents a unit vector from the following vectors is .....

(a) $(-3, 2, 2)$ (b) $\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}\right)$

(c) $\left(\frac{3}{5}, \frac{-4}{5}, 0\right)$ (d) $\left(\frac{-1}{12}, \frac{-3}{12}, \frac{2}{12}\right)$

**(12)** *(2nd Session 2023)* If $\vec{A} = (-3, 0, 4)$ , then the direction cosines of the vector $\vec{A}$ are .....

(a) $\left(\frac{3}{5}, 0, \frac{-4}{5}\right)$ (b) $\left(\frac{3}{5}, 0, \frac{4}{5}\right)$

(c) $\left(\frac{-3}{5}, 0, \frac{4}{5}\right)$ (d) $\left(\frac{-3}{5}, 0, \frac{-4}{5}\right)$

**(13)** *(2nd Session 2017)* If the direction angles of a vector are $(45°, 45°, \theta)$ , then $\theta$ = .....

(a) 45° (b) 90° (c) 0° (d) 60°

**(14)** *(2nd Session 2024)* If the measures of direction angles of vector $\vec{A}$ are $(90° , 60° , \theta)$ , then one of the values of $\theta$ is .....

(a) $90°$ (b) $120°$ (c) $135°$ (d) $150°$

**(15)** If $\| 4 \vec{A} \| = \| 3 k \vec{A} \|$ , then k = .....

(a) $\pm 1$ (b) $\pm \frac{3}{4}$ (c) $\pm \frac{4}{3}$ (d) $\pm \frac{2}{\sqrt{3}}$

**(16)** If $\vec{A} = (1 , 2 , -4)$ , $\vec{B} = (1 , 1 , k - 1)$ and $\| \vec{A} + \vec{B} \| = 7$ units , then k = .....

(a) $\pm 6$ (b) 2 or $-11$ (c) 11 or $-1$ (d) $\pm 7$

**(17)** If $\overrightarrow{AB} = (2 , 5 , -1)$ , $\vec{A} = (2 , 3 , 1)$ , then $\vec{B}$ = .....

(a) $(4 , 8 , 0)$ (b) $(4 , 0 , 8)$ (c) $(0 , 4 , 8)$ (d) $\vec{O}$

**(18)** If $\vec{A} = \left( \frac{-1}{2} , \frac{3}{4} , k \right)$ is a unit vector , then the value of k = .....

(a) $\pm 3$ (b) $\pm\sqrt{3}$ (c) $\pm \frac{\sqrt{3}}{2}$ (d) $\pm \frac{\sqrt{3}}{4}$

**(19)** The vector $\vec{A} = 3 \hat{i} + \hat{j} - 2 \hat{k}$ makes an angle of measure ..... (to the nearest degree) with the positive direction of $\mathcal{X}$-axis.

(a) $74°$ (b) $78°$ (c) $143°$ (d) $37°$

**(20)** The vector $\vec{B} = \hat{i} + 2 \hat{j}$ makes an angle of measure ..... with the positive direction of z-axis.

(a) zero (b) $90°$ (c) $180°$ (d) $270°$

**(21)** If the measure of the angle which $\vec{C} = (2 , 4 , k)$ makes with the positive direction of y-axis equals $45°$ , then k = .....

(a) $\pm 5$ (b) $\pm 2$ (c) $\pm 2\sqrt{3}$ (d) $\pm\sqrt{3}$

**(22)** If the measures of direction angles of position vector $\vec{A}$ are $(30° , 120° , 90°)$ , then $\vec{A}$ lies in the .....

(a) $\mathcal{X}$ z-plane. (b) y z-plane. (c) $\mathcal{X}$ y-plane. (d) $\overrightarrow{O\mathcal{X}}$ direction.

**(23)** The measures of the direction angles of the positive y-axis are .....

(a) $(90° , 90° , 0)$ (b) $(0 , 90° , 90°)$ (c) $(0 , 90° , 0)$ (d) $(90° , 0 , 90°)$

**(24)** The direction cosines of the z-axis are .....

(a) $(0 , 0 , 1)$ (b) $(0 , 1 , 0)$ (c) $(0 , 0 , -1)$ (d) $(0 , -1 , 0)$

**(25)** The norm of a vector is 6 units and its direction cosines are $\left(-\frac{1}{2}, 0, \frac{\sqrt{3}}{2}\right)$, then the vector is .....

(a) $\left(-3, 0, 3\sqrt{3}\right)$ (b) $\left(-3, 3, 3\sqrt{3}\right)$

(c) $\left(\frac{-3}{2}, 0, 2\sqrt{3}\right)$ (d) $\left(-1, 0, \sqrt{3}\right)$

**(26)** If $(60°, \theta_y, 45°)$ are the measures of direction angles of a vector then $\theta_y$ = .....

(a) 30° or 60° (b) 60° or 150° (c) 60° or 120° (d) 30° or 150°

**(27)** If the direction cosines of a vector are $\left(\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$, then its direction angles are .....

(a) (60° , 120° , 135°) (b) (60° , 120° , 45°)

(c) (30° , 150° , 45°) (d) (120° , 60° , 45°)

**(28)** If the measures of direction angles of vector $\vec{A}$ are (45° , 135° , 90°) , then the unit vector in direction of $\vec{A}$ = .....

(a) $\left(\sqrt{2}, -\sqrt{2}, 0\right)$ (b) $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 1\right)$ (c) $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right)$ (d) $(1, -1, 0)$

**(29)** If the measures of direction angles of vector $\vec{A}$ are (45° , 120° , 60°) , $\|\vec{A}\| = 12\sqrt{2}$ , then $\vec{A}$ = .....

(a) $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, \frac{1}{2}\right)$ (b) $\left(12, -6\sqrt{2}, 6\sqrt{2}\right)$

(c) $\left(12, 6\sqrt{2}, -6\sqrt{2}\right)$ (d) $\left(\frac{1}{\sqrt{2}}, \frac{-1}{2}, \frac{-1}{2}\right)$

**(30)** If $\vec{A} = \left(2\sqrt{2}, 2, -2\right)$, then the vector which have the same direction angles is .....

(a) $\left(8, 0, -4\sqrt{3}\right)$ (b) $\left(4\sqrt{2}, -4, -4\right)$

(c) $\left(3\sqrt{2}, 3, -3\right)$ (d) $\left(4, 2, \sqrt{3}\right)$

**(31)** The unit vector of a vector parallel to the y z-plane could be .....

(a) $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ (b) $\left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (c) $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$ (d) $\left(\frac{1}{\sqrt{2}}, 0, 0\right)$

**(32)** If the vector $\vec{n} = (a, 4, c)$ is parallel to the cartesian plane y z and $\|\vec{n}\| = 5$ , then $c^2$ = .....

(a) 3 (b) 9 (c) 12 (d) 20

**(33)** The vector which makes equal angles with the positive direction of the coordinate axes from the following is .....

(a) $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ (b) $\left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$

(c) $\left(-\sqrt{3}, -\sqrt{3}, \sqrt{3}\right)$ (d) $(-1, 1, -1)$

**(34)** If C is the midpoint of $\overline{AB}$ and $\overrightarrow{AC} = (3, -2, 5)$, then $\overrightarrow{CB}$ = .....

(a) $(-3, 2, -5)$ (b) $(3, -2, 5)$ (c) $\left(\frac{-3}{2}, 1, \frac{-5}{2}\right)$ (d) $\left(\frac{3}{2}, -1, \frac{5}{2}\right)$

**(35)** If $\vec{A}$ is a non-zero vector, $\vec{C} = \frac{\vec{A}}{\|\vec{A}\|}$, then which of the following statements is always true ?

(a) $\|\vec{A}\| = \|\vec{C}\|$ (b) $\|\vec{A}\| < \|\vec{C}\|$ (c) $\|\vec{A}\| > \|\vec{C}\|$ (d) $\vec{A} \,//\, \vec{C}$

**(36)** If $\vec{A}, \vec{B}, \vec{C}$ are three vectors, then $\left\|\frac{\vec{A}}{\|\vec{A}\|}\right\| + 2\left\|\frac{\vec{B}}{\|\vec{B}\|}\right\| + 3\left\|\frac{\vec{C}}{\|\vec{C}\|}\right\|$ = .....

(a) 4 (b) 5 (c) 6 (d) 8

**(37)** If $\vec{A} = (4, 0, 3)$, then tangent of the angle this vector makes with the $\mathcal{X}$-axis = .....

(a) $\frac{3}{4}$ (b) $\frac{4}{5}$ (c) $\frac{4}{\sqrt{7}}$ (d) $\frac{4}{7}$

**(38)** ***(1st Session 2021)*** If cosine of the angle between the vector $\vec{A} = (k, 12, 4)$ and the $\mathcal{X}$-axis equals $\frac{3}{13}$, then k = ..... where $k \in \mathbb{R}$

(a) 4 (b) $\sqrt{3}$ (c) $-\sqrt{3}$ (d) 3

**(39)** The position vector which lies in the positive coordinate plane $\mathcal{X}z$ and makes an angle of measure 30° with the positive $\mathcal{X}$-axis its direction cosines are .....

(a) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$ (b) $\left(\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right)$

(c) $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$ (d) $\left(\frac{\sqrt{3}}{2}, 0, -\frac{1}{2}\right)$

**(40)** The position vector that lies on the $\mathcal{X}$ y-plane and makes an angle of measure 60° with the positive direction of y-axis, then its direction cosines are .....

(a) $\left(0, \frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ (b) $\left(\frac{1}{2}, \pm\frac{\sqrt{3}}{2}, 0\right)$

(c) $\left(\pm\frac{\sqrt{3}}{2}, \frac{1}{2}, 0\right)$ (d) $\left(\frac{\sqrt{3}}{2}, \pm\frac{1}{2}, 0\right)$

**(41)** If $\|\vec{A}\| = 3$, $-1 \le k \le 2$, then $\|k\vec{A}\| \in$ .....

(a) $[-3, 6]$ (b) $[0, 6]$ (c) $[3, 6]$ (d) $[1, 6]$

**(42)** The vector whose direction angles are $(0 , 90° , 90°)$ .....

(a) acts in the positive direction of y-axis.

(b) acts in the positive direction of $\mathcal{X}$-axis.

(c) acts in the positive direction of z-axis.

(d) lies in the y z-plane.

**(43)** Which of the following could be direction angles of vector in space ?

(a) $(60° , 30° , 30°)$ (b) $(90° , 90° , 60°)$

(c) $(120° , 150° , 90°)$ (d) $(60° , 30° , 0°)$

**(44)** All of the following represents unit vectors except .....

(a) $\left(\frac{1}{\sqrt{2}} , 0 , \frac{-1}{\sqrt{2}}\right)$ (b) $\left(\frac{-6}{10} , \frac{2\sqrt{7}}{10} , 0.6\right)$

(c) $(1 , 1 , 1)$ (d) $\left(\frac{1}{2} , \frac{-\sqrt{3}}{2} , 0\right)$

**(45)** If K , E , F are direction cosines of vector $\overrightarrow{A}$ , then .....

(a) $K + E + F = 1$ (b) $K = E = F$

(c) $K^2 + E^2 + F^2 = 1$ (d) $K + E + F = \| \overrightarrow{A} \|$

**(46)** *(2nd Session 2021)* If $\overrightarrow{A} + \overrightarrow{BC} = 4\hat{i} + 12\hat{j} + 9\hat{k}$ , where $\overrightarrow{A} = (0 , -1 , 3)$ , $B (4 , -2 , 1)$ , then $\overrightarrow{C}$ = .....

(a) $8\hat{i} + 13\hat{j} + 13\hat{k}$ (b) $8\hat{i} + 11\hat{j} + 7\hat{k}$

(c) $8\hat{i} + 9\hat{j} + 7\hat{k}$ (d) $8\hat{i} + 13\hat{j} - 7\hat{k}$

**(47)** *(2nd Session 2021)* The direction cosines of the vector $\overrightarrow{A} = (-2k , 2k , k)$ , where $k \in ]0 , 1[$ are .....

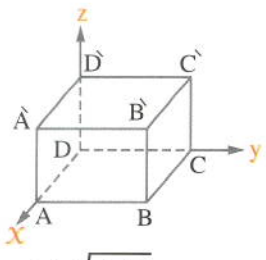
(a) $\left(\frac{-2}{3} , \frac{2}{3} , \frac{1}{3}\right)$ (b) $\left(\frac{-2k}{3} , \frac{2k}{3} , \frac{-k}{3}\right)$

(c) $\left(\frac{-2}{3} , \frac{-2}{3} , \frac{-1}{3}\right)$ (d) $\left(\frac{2k}{3} , \frac{2k}{3} , \frac{k}{3}\right)$

**(48)** *(1st Session 2022)* If $A (3 , 2 , 5)$ is a point in space , its projection on $\mathcal{X}$ y-plane is point B and its projection on $\mathcal{X}$ z-plane is point C , then $\overrightarrow{BC}$ = .....

(a) $-2\hat{i} + 3\hat{j}$ (b) $2\hat{j} - 5\hat{k}$ (c) $-2\hat{j} + 5\hat{k}$ (d) $-2\hat{i} - 3\hat{k}$

**(49) In the opposite figure :**



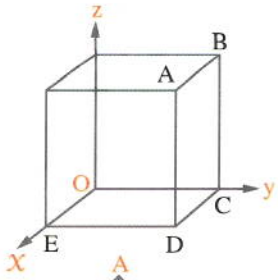

ABCD $\grave{A}\grave{B}\grave{C}\grave{D}$ is a cuboid , $A (4 , 0 , 0)$ ,

$C (0 , 9 , 0)$ , $\grave{D} (0 , 0 , 7)$

, then $\| \overrightarrow{A\grave{C}} \|$ = ..... length unit.

(a) $\sqrt{137}$ (b) $\sqrt{141}$ (c) $\sqrt{146}$ (d) $\sqrt{147}$

**(50)** The opposite figure represents a cuboid , A (6 , 8 , 24) , then the direction angles of the vector $\overrightarrow{OD}$ are .....




(a) (60° , 120° , 45°)

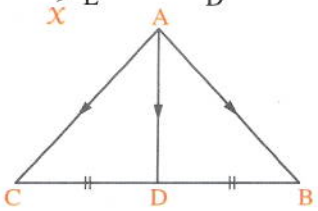
(b) (53° 8` , 36° 52` , 90°)

(c) (53° 8` , 90° , 36° 52`)

(d) (36° 52` , 45° , 90°)

**(51)** *(Speciman 2025)* **In the opposite figure :**

If D is the midpoint of $\overline{BC}$ and $\overrightarrow{AB} = 5\hat{i} - 2\hat{j} + 7\hat{k}$ , $\overrightarrow{AC} = 3\hat{i} - 2\hat{j} + \hat{k}$ , then $\| \overrightarrow{AD} \|$ = .....




(a) 5 (b) 6 (c) 7 (d) 8

**(52)** If (30° , 60° , 90°) are the direction angles of vector $\vec{A}$ , then the direction angles of vector $2\vec{A}$ is .....

(a) (30° , 60° , 90°) (b) (60° , 120° , 180°)

(c) (– 30° , – 60° , – 90°) (d) (60° , 120° , 90°)

**(53)** If (60° , 135° , 60°) are the direction angles of vector $\vec{A}$ , then the direction angles of vector $-2\vec{A}$ is .....

(a) (60° , 135° , 60°) (b) (120° , 270° , 120°)

(c) (30° , 45° , 30°) (d) (120° , 45° , 120°)

**(54)** Which of the following vectors represents a perpendicular unit vector on the cartesian plane $x$ y ?

(a) $\hat{i}$ (b) $\hat{j}$ (c) $\hat{k}$ (d) (0 , 0 , 2)

**(55)** If $(\theta_x , \theta_y , \theta_z)$ are the direction angles of a position vector where $\theta_x + \theta_y = 90°$ , then which of the following not correct ?

(a) $\theta_z = 90°$

(b) The vector lies in the plane $x$ y

(c) $\cos^2 \theta_x + \cos^2 \theta_y = 1$

(d) The vector makes equal angles with the coordinate axes.

**(56)** If $(\theta_x , \theta_y , \theta_z)$ are direction angles of the vector $\vec{A}$ in 3-diamension space , then which of the following is incorrect ?

**(1)** $\theta_x + \theta_y \geq 90°$ **(2)** $\sin^2\left(\frac{\pi}{2} - \theta_x\right) + \sin^2\left(\frac{\pi}{2} - \theta_y\right) + \sin^2\left(\frac{\pi}{2} - \theta_z\right) = 1$

**(3)** $(-\theta_x , -\theta_y , -\theta_z)$ are direction angles of the vector $-\vec{A}$

**(4)** $\theta_x = \pm \cos^{-1}\left(\frac{a_x}{\| \vec{A} \|}\right)$

(a) **(1)** only. (b) **(1)** , **(2)** only. (c) **(2)** , **(3)** only. (d) **(3)** , **(4)** only.

**8** Find the value of : L , m , n that makes the two vectors $\vec{A} = (L-4 , m^2-3 , 1)$ and $\vec{B} = (5 , 1 , n^2)$ are equal. « 9 , ± 2 , ± 1 »

**9** If $= (2x+1 , 5 , k+4) = (-1 , y^2-4 , x+1)$ find the value of : $x$ , y and k « – 1 , ± 3 , – 4 »

**10** **Find the values of $\ell$ , m , n in each of the following :**

**( 1 )** $2(3 , m , \ell) + 3(n , -2 , 4) = (3 , 2 , 4)$

**( 2 )** $\ell(-6 , m , 2n) = 12\hat{i} + 10\hat{i} - 4\hat{k}$

**( 3 )** $\ell(-2 , 3 , 1) + m(-3 , 1 , 2) + n\hat{k} = (-8 , 5 , 8)$

**11** If $\vec{A} = (4 , 3 , 2)$ , $\vec{B} = (-2 , 5 , -1)$ , $\vec{C} = (8 , 19 , 4)$

**( 1 )** Find in terms of fundamental unit vectors each of : $\vec{B} - \vec{C}$ , $2(\vec{B} + \vec{C})$

**( 2 )** Find : $\vec{C}$ in terms of $\vec{A}$ and $\vec{B}$

**12** If $\vec{A} = (-3 , 2 , 1)$ , $\vec{B} = (2 , 3 , 0)$ Find : $\|\vec{A}\| , \|\vec{A} + \vec{B}\|$ « $\sqrt{14} , 3\sqrt{3}$ »

**13** If $\vec{A} = (4 , -6 , 2)$ , $3(\vec{A} - \vec{B}) = 7\vec{B} - 2\vec{A}$ Find : $\|\vec{B}\|$ « $\sqrt{14}$ »

**14** If A $(-1 , 4 , 6)$ , B $(2 , 5 , -1)$ Find : $\|\overrightarrow{AB}\|$ « $\sqrt{59}$ »

**15** If $K\|4\vec{A}\| = \|-3\vec{A}\|$ Find the value of k where the vector $\vec{A}$ is a vector in 3-D space , $K \in R^*$ « $\frac{3}{4}$ »

**16** If $\|-8\vec{A}\| = 5\|K\vec{A}\|$ Find the value of k where $\vec{A}$ is a vector in 3-D space , $K \in R^*$ « $\pm\frac{8}{5}$ »

**17** If $\vec{A} = (k , 0 , 0)$ , $\hat{i} = (1 , 0 , 0)$ Prove that : $\|\vec{A}\| = |k|\|\hat{i}\|$

**18** If $\vec{A} = (3 , -2 , 4)$ , $\vec{B} = (-2 , 4 , 2)$ , $\vec{C} = (7 , 1 , -3)$ **Find :**

**( 1 )** $\|\vec{A} + 2\vec{C}\|$ **( 2 )** $\|5\vec{B} + 3\vec{A}\|$

**( 3 )** $\|\vec{A} + \frac{1}{2}\vec{B}\|$ **( 4 )** $\|\vec{A} + \vec{B} - 2\vec{C}\|$

**( 5 )** $\|2\vec{A} + 3\vec{B} - 4\vec{C}\|$ « $\sqrt{293} , \sqrt{681} , \sqrt{29} , \sqrt{313} , 6\sqrt{41}$ »

**19** If $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$ , $\vec{B} = -\hat{i} - 2\hat{j}$ **Find :**

**( 1 )** $2\vec{A} - 3\vec{B}$

**( 2 )** $\|\vec{A} + \vec{B}\| , \|\vec{A}\| + \|\vec{B}\|$ What do you notice ?

**20** If $\vec{A}$ and $\vec{B}$ are two vectors in $\mathbb{R}^3$. Is $\|\vec{A} + \vec{B}\| = \|\vec{A}\| + \|\vec{B}\|$ ?
Which side is greater if both sides are unequal ?

**21** **Find the unit vector in direction of each of the following vectors :**

( 1 ) $\vec{A} = (8 , -4 , -8)$

( 2 ) $\vec{B} = \hat{i} - 2\hat{j} - \hat{k}$

( 3 ) $\vec{C} = 3\hat{i} - 4\hat{k}$

( 4 ) $\vec{D} = (2 , -1 , 3)$

---

**22** If $\vec{A} = (-1 , 3 , 2)$ , $\vec{B} = (4 , 2 , -1)$ **Find the unit vector in direction of :**

( 1 ) $\vec{A}$

( 2 ) $\vec{B}$

( 3 ) $\overrightarrow{AB}$

( 4 ) $\overrightarrow{BA}$

---

**23** If $\vec{C} = \left(\frac{1}{\sqrt{2}} , -\frac{1}{2} , \frac{1}{2}\right)$ is a unit vector in direction of each $\vec{A}$ and $\vec{B}$ and $\| 3\vec{A} \| = \| -5\,B \| = 15$

Find each of : $\vec{A}$ and $\vec{B}$

---

**24** If $\vec{A}$ is a unit vector where $\vec{A} = \left(k , \frac{2}{3} , \frac{2}{3}\right)$ Find : k « $\pm\frac{1}{3}$ »

---

**25** If the two vectors : $\vec{B} = \left(L , 4\,m , \frac{1}{5}\right)$ , $\vec{C} = \frac{-2\sqrt{2}}{5}\hat{i} + \frac{4}{5}\hat{j} + \frac{n}{5}\hat{k}$ are unit vectors in direction of $\vec{A}$

( 1 ) Find : L , m , n

( 2 ) If $\| \vec{A} \| = 15$ Find : $\vec{A}$

---

**26** **Find the direction cosines of each of the following vectors :**

( 1 ) $\vec{A} = 5\hat{i} - 2\hat{j} + \sqrt{7}\hat{k}$

( 2 ) $\vec{B} = \left(-2 , 2\sqrt{2} , 2\right)$

( 3 ) $\vec{C} = 4\hat{i} - 3\hat{k}$

( 4 ) $\vec{D} = -5\hat{j} + 12\hat{k}$

---

**27** **Find the measures of direction angles of each of the following vectors :**

( 1 ) $\vec{A} = (8 , 6 , -10)$

( 2 ) $\vec{C} = -2\hat{i} + 3\hat{j} - 6\hat{k}$

( 3 ) $\vec{E} = \hat{k}$

( 4 ) $\vec{M} = \hat{i} + \hat{j}$

---

**28** If the vector $\vec{A}$ makes with the (+ve) direction of the coordinate axes $x$ , y , z angles of measure 60° , 80° , θ° where θ is an acute angle.

( 1 ) Find value of θ

( 2 ) Write the cartesian form of $\vec{A}$ if $\| \vec{A} \| = 13$

---

**29** Mention the direction angles and direction cosines of each of the fundamental axes in 3-D space.

---

**30** Find the direction angles and direction cosines of a vectors makes equal angles with the positive coordinate axes.

**31** Find the direction angles of vector $\vec{A}$ which lies in the coordinate plane y z and makes angle of measure 15° with y-axis.

**32** *(1st Session 2017)* Find the algebraic form of vector $\vec{A}$ whose norm is $21\sqrt{3}$ and makes equal angles with the positive coordinate axes.

**33** The vector $\vec{A}$ makes angle of measure 30° with positive $x$-axis and 60° with positive y-axis. Find the measure of the angle it makes with the positive z-axis.

**34** What can you say about the coordinates of the vector $\vec{A}$ if the vector $\vec{A}$ is parallel to y z-plane ?

**35** If $\theta_x$ , $\theta_y$ , $\theta_z$ are the direction angles of a vector with the positive coordinate axes.

Prove that : $\sin^2\theta_x + \sin^2\theta_y + \sin^2\theta_z = 2$

**36** **In the opposite figure :**

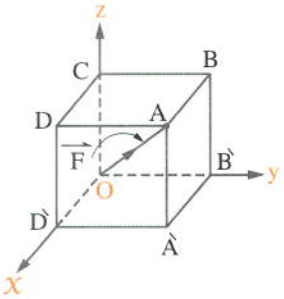
A cube of edge length 5 length unit.

The magnitude of the force $\vec{F}$ is 25 newton.

**Find :** ( 1 ) The direction cosines of the vector $\vec{F}$

( 2 ) The measures of the direction angles of vector $\vec{F}$

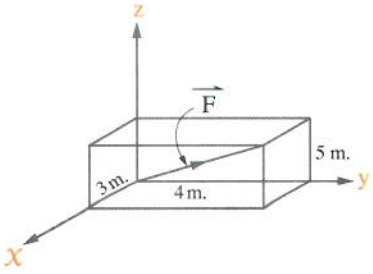
( 3 ) The vector $\vec{F}$ in terms of fundamental unit vectors in the 3 - dimension space.




**37** The opposite figure represents the force $\vec{F}$ with a magnitude 200 Newton

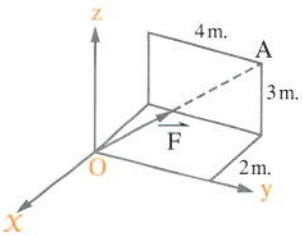
( 1 ) Express the force $\vec{F}$ in an algebraic form.

( 2 ) Find the measure of the direction angles of the force $\vec{F}$



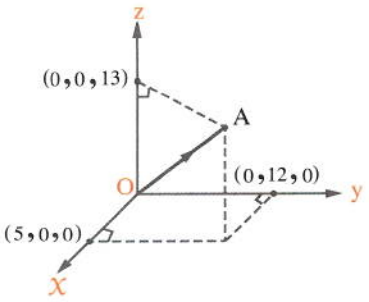

**38** **In the opposite figure :**

Find the components of the force $\vec{F}$ whose magnitude is $12\sqrt{29}$ newtons in the directions of the coordinate axes.



**39** **In the opposite figure :**

Find the directions cosines of vector $\overrightarrow{OA}$

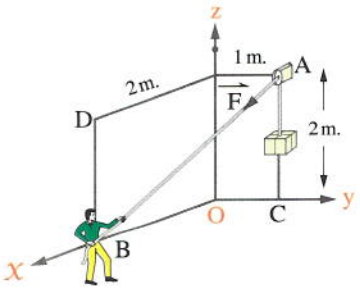



**40** **In the opposite figure :**

If the tension force in a string equals 21 newtons ,

find the algebraic components of the forces $\vec{F}$

in the directions of the cartesian axes.

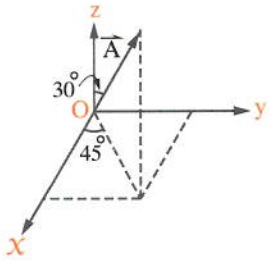



**41** **The opposite figure :**

Represents the vector $\vec{A}$ of norm 10 units

( 1 ) Express the vector $\vec{A}$ in algebraic form (Cartesian components)

( 2 ) Find the measures of direction angles of the vector $\vec{A}$




**42** **In the opposite figure :**

( 1 ) Find $\| \vec{A} \|$

( 2 ) Find $\vec{A}$ in terms of fundamental unit vectors in the 3-D space.

( 3 ) Find the measures of direction angles of vector $\vec{A}$

